

# Introduction to Artificial Intelligence

Lecture 4: Quantifying uncertainty

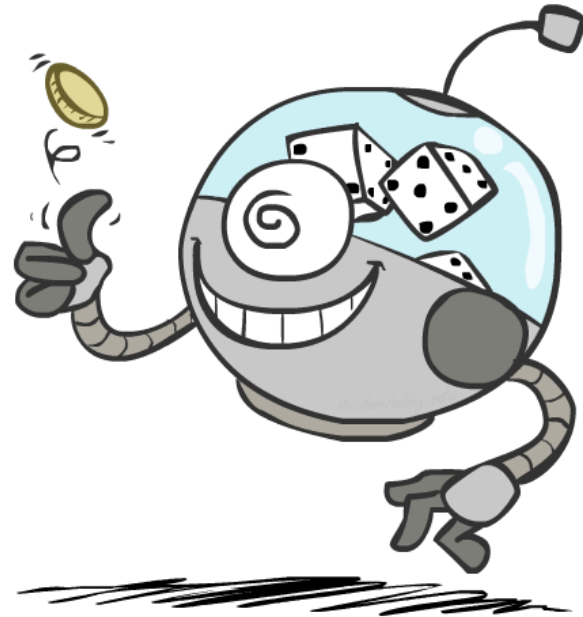
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# Today

- Random variables
- Probability distributions
- Inference
- Independence
- The Bayes' rule



Do not overlook this lecture!



## Principle of maximum expected utility

From Lecture 1, a rational agent chooses the action of **highest expected utility**, given what it has perceived,

$$a_t = \arg \max_{a \in \mathcal{A}} \mathbb{E} [V(s_{1:T}) \mid e_{1:t}, a_t = a],$$

where  $V : \mathcal{S}^* \rightarrow \mathbb{R}$  scores the states  $s_{1:T}$ , and the average is over the states the agent may reach, following  $P(s_{t+1} \mid s_t, a_t)$  and  $P(e_t \mid s_t)$ .

What are these probabilities, and how do we reason with them?

# Quantifying uncertainty

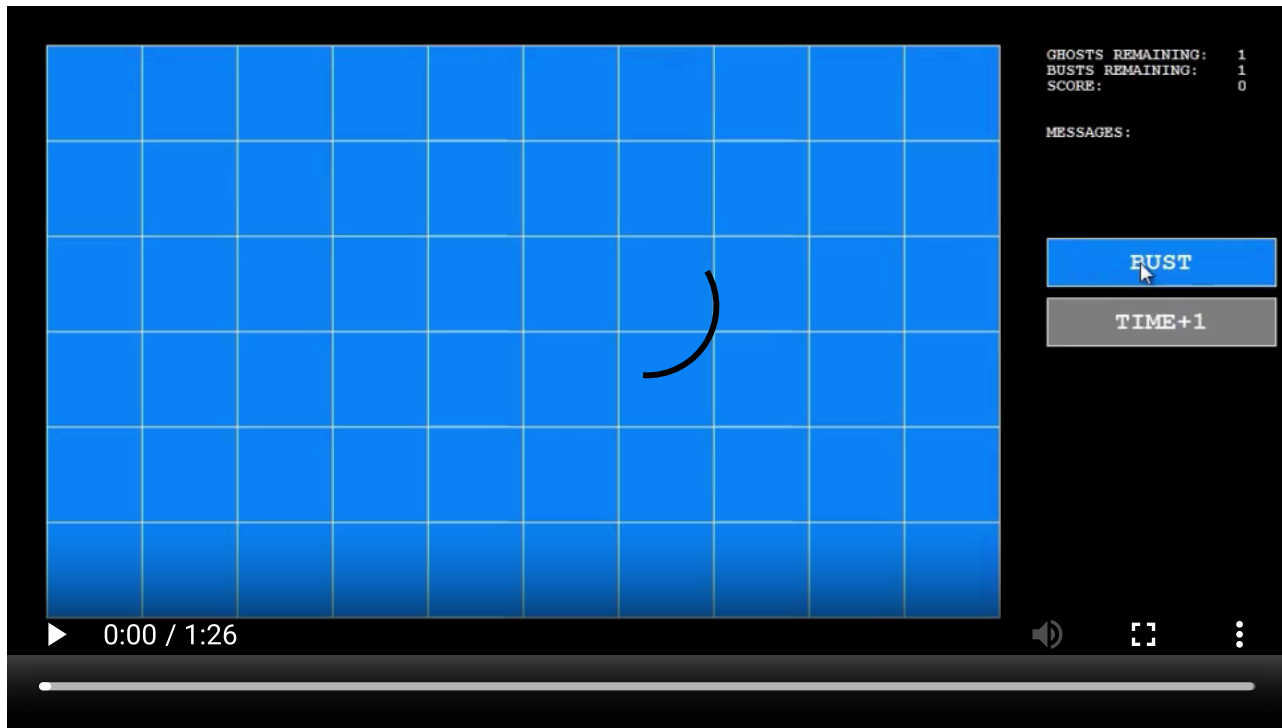
A ghost is **hidden** in the grid somewhere.

Sensor readings tell how close a square is to the ghost:

- On the ghost: red
- 1 or 2 away: orange
- 3 away: yellow
- 4+ away green



Sensors are **noisy**, but we know the probability values  $P(\text{color} \mid \text{distance})$ , for all colors and all distances.



# Uncertainty

General setup:

- **Observed** variables or evidence: agent knows certain things about the state of the world (e.g., sensor readings).
- **Unobserved** variables: agent needs to reason about other aspects that are uncertain (e.g., where the ghost is).
- (Probabilistic) **model**: agent knows or believes something about how the observed variables relate to the unobserved variables.

**Probabilistic reasoning** provides a framework for managing our knowledge and beliefs.

# Probabilistic assertions

A **probabilistic assertion** attaches a number to a proposition,

$$P(\text{ghost in cell } [3, 2]) = 0.02,$$

as a degree of belief, not as a claim about the truth of the proposition.

The agent holds beliefs rather than certainties for two reasons:

- **ignorance**: it does not observe the state, and its models of the world are approximate;
- **laziness**: listing every case and every exception would cost more than it is worth.

## Frequentism vs. Bayesianism

What does a probability value represent? Take  $P(\text{ghost in cell } [3, 2]) = 0.02$ .

- The objectivist, **frequentist** view: a property of the world, the fraction of games in which the ghost sits in that cell.
- The subjectivist, **Bayesian** view: the degree of belief of the agent that the ghost is in that cell, given the readings it has taken.

The agent plays a single game, with no long run to count, so we read probabilities as beliefs in this course: a new reading changes the number, not the ghost.

# Kolmogorov's axioms

Begin with a set  $\Omega$ , the **sample space**. Its elements  $\omega \in \Omega$  are the **sample points**, or possible worlds.

A **probability space** is a sample space equipped with a probability function  $P$  that assigns a number  $P(\omega)$  to each sample point, such that

- $0 \leq P(\omega) \leq 1$  for all  $\omega \in \Omega$ ;
- $\sum_{\omega \in \Omega} P(\omega) = 1$ .

## Events

An **event** is a set of sample points  $A \subseteq \Omega$ , and its probability is the mass it collects,

$$P(A) = \sum_{\omega \in A} P(\omega).$$

Equivalently, in Kolmogorov's terms,

- $P(A) \geq 0$  for all events;
- $P(\Omega) = 1$ ;
- $P(A \cup B) = P(A) + P(B)$  for disjoint events.

It follows that

$$P(\neg a) = 1 - P(a), \quad P(a \vee b) = P(a) + P(b) - P(a \wedge b).$$

### Example

- $\Omega$  = the 6 possible rolls of a die.
- $\omega_i$  (for  $i = 1, \dots, 6$ ) are the sample points, each corresponding to an outcome of the die.
- Assignment  $P$  for a fair die:

$$P(1) = P(2) = P(3) = P(4) = P(5) = P(6) = \frac{1}{6}$$

- "3 or more" is the event  $A = \{3, 4, 5, 6\}$ , of probability

$$P(A) = P(3) + P(4) + P(5) + P(6) = \frac{2}{3},$$

hence  $P(\neg A) = 1 - P(A) = \frac{1}{3}$  for "2 or less".

# Random variables

- A **random variable** is a function  $X : \Omega \rightarrow D_X$  from the sample space to a domain  $D_X$  of outcomes.
- " $X = x$ " is the event  $\{\omega \in \Omega : X(\omega) = x\}$ , so  $P$  induces a **probability distribution** over  $X$ ,

$$P(X = x) = \sum_{\{\omega : X(\omega) = x\}} P(\omega).$$

- The events we care about are the **assignments** of random variables, such as  $X_1 = x_1, \dots, X_n = x_n$ .

*Example:*

On the die, "odd" is the random variable

$$\text{Odd} : \Omega \rightarrow \{\text{true}, \text{false}\}, \quad \text{Odd}(\omega) = (\omega \bmod 2 = 1),$$

whose distribution is

$$P(\text{Odd} = \text{true}) = P(1) + P(3) + P(5) = \frac{1}{2}, \quad P(\text{Odd} = \text{false}) = \frac{1}{2}.$$

## Notations

- Random variables are named in upper case,  $X$ ,  $Y$ ,  $Weather$ ; their values in lower case,  $x$ ,  $y$ ,  $sun$ .
- $D_X$  is the domain of  $X$ , so that  $x \in D_X$  is one of its values.
- $P(X = x)$  is a **number**, abbreviated  $P(x)$  when clear from context.
- $\mathbf{P}(X)$  is the **distribution** of  $X$ , i.e. the table of the numbers  $P(X = x)$  for all  $x \in D_X$ , and not a single value.

# Probability distributions

## Probability mass function

For a discrete variable  $X$ , the probability distribution  $\mathbf{P}(X)$  is the list of the values  $P(X = x)$  for all  $x \in D_X$ , known as the **probability mass function**.

$\mathbf{P}(W)$

| $W$    | $P$ |
|--------|-----|
| sun    | 0.6 |
| rain   | 0.1 |
| fog    | 0.3 |
| meteor | 0.0 |



## Joint distributions

A **joint** probability distribution over a set of random variables  $X_1, \dots, X_n$  specifies the probability of each (combined) outcome:

$$P(X_1 = x_1, \dots, X_n = x_n) = \sum_{\{\omega: X_1(\omega)=x_1, \dots, X_n(\omega)=x_n\}} P(\omega)$$

$$\mathbf{P}(T, W)$$

| $T$  | $W$  | $P$ |
|------|------|-----|
| hot  | sun  | 0.4 |
| hot  | rain | 0.1 |
| cold | sun  | 0.2 |
| cold | rain | 0.3 |

Over  $n$  boolean variables, the table has  $2^n$  entries, hence  $2^n - 1$  free numbers.

## Marginal distributions

The **marginal distribution** of a subset of random variables is obtained by summing the joint distribution over the values of the other variables (the **sum rule**),

$$P(x) = \sum_y P(x, y), \quad \mathbf{P}(X) = \sum_y \mathbf{P}(X, Y = y).$$

$\mathbf{P}(T, W)$

| $T$  | $W$  | $P$ |
|------|------|-----|
| hot  | sun  | 0.4 |
| hot  | rain | 0.1 |
| cold | sun  | 0.2 |
| cold | rain | 0.3 |

$\mathbf{P}(T)$

| $T$  | $P$ |
|------|-----|
| hot  | 0.5 |
| cold | 0.5 |

$\mathbf{P}(W)$

| $W$  | $P$ |
|------|-----|
| sun  | 0.6 |
| rain | 0.4 |

$$P(t) = \sum_w P(t, w)$$

$$P(w) = \sum_t P(t, w)$$

Intuitively, marginal distributions are sub-tables which eliminate variables.

## Conditional distributions

The **conditional probability** of  $a$  given  $b$ , for  $P(b) > 0$ , is the ratio of the probability of  $a$  and  $b$  to the probability of  $b$ ,

$$P(a \mid b) = \frac{P(a, b)}{P(b)}.$$

Indeed, observing  $B = b$  rules out all those possible worlds where  $B \neq b$ , leaving a set whose total probability is just  $P(b)$ . Within that set, the worlds for which  $A = a$  satisfy  $A = a \wedge B = b$  and constitute a fraction  $P(a, b)/P(b)$ .



Conditional distributions are probability distributions over some variables, given **fixed** values for others. Writing  $\mathbf{P}(X \mid Y)$  denotes them all at once: one distribution  $\mathbf{P}(X \mid Y = y)$  for each value  $y$  of  $Y$ .

$$\mathbf{P}(T, W)$$

| $T$  | $W$  | $P$ |
|------|------|-----|
| hot  | sun  | 0.4 |
| hot  | rain | 0.1 |
| cold | sun  | 0.2 |
| cold | rain | 0.3 |

$$\mathbf{P}(W \mid T = \text{hot})$$

| $W$  | $P$ |
|------|-----|
| sun  | 0.8 |
| rain | 0.2 |

$$\mathbf{P}(W \mid T = \text{cold})$$

| $W$  | $P$ |
|------|-----|
| sun  | 0.4 |
| rain | 0.6 |

## Product rule

Rearranging the definition of the conditional probability,

$$P(a, b) = P(a \mid b)P(b), \quad \mathbf{P}(A, B) = \mathbf{P}(A \mid B)\mathbf{P}(B),$$

the second form standing for one equation per value of  $A$  and  $B$ .

Summing it over the values of  $B$  gives the **law of total probability**,

$$P(a) = \sum_b P(a \mid b)P(b).$$

$\mathbf{P}(W)$

| $W$  | $P$ |
|------|-----|
| sun  | 0.8 |
| rain | 0.2 |

$\mathbf{P}(D \mid W)$

| $D$ | $W$  | $P$ |
|-----|------|-----|
| wet | sun  | 0.1 |
| dry | sun  | 0.9 |
| wet | rain | 0.7 |
| dry | rain | 0.3 |

$\mathbf{P}(D, W)$

| $D$ | $W$  | $P$ |
|-----|------|-----|
| wet | sun  | ?   |
| dry | sun  | ?   |
| wet | rain | ?   |
| dry | rain | ?   |

## Chain rule

More generally, any joint distribution can always be written as an incremental product of conditional distributions:

$$P(x_1, x_2, x_3) = P(x_1)P(x_2 \mid x_1)P(x_3 \mid x_1, x_2)$$
$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i \mid x_1, \dots, x_{i-1})$$

In the same way,  $\mathbf{P}(X_1, \dots, X_n) = \prod_{i=1}^n \mathbf{P}(X_i \mid X_1, \dots, X_{i-1})$ .

## Normalization trick

$$\mathbf{P}(T, W)$$

| $T$  | $W$  | $P$ |
|------|------|-----|
| hot  | sun  | 0.4 |
| hot  | rain | 0.1 |
| cold | sun  | 0.2 |
| cold | rain | 0.3 |

$$\rightarrow \mathbf{P}(T = \text{cold}, W)$$

| $T$  | $W$  | $P$ |
|------|------|-----|
| cold | sun  | 0.2 |
| cold | rain | 0.3 |

**Select** the joint probabilities matching the evidence  $T = \text{cold}$ .

$$\rightarrow \mathbf{P}(W \mid T = \text{cold})$$

| $W$  | $P$ |
|------|-----|
| sun  | 0.4 |
| rain | 0.6 |

**Normalize** the selection (make it sum to **1**).

# Inference

What an agent asks, given what it has observed:

$$P(\text{on time} \mid \text{no reported accidents}) = 0.9$$

$$P(\text{on time} \mid \text{no reported accidents, 5AM}) = 0.95$$

$$P(\text{on time} \mid \text{no reported accidents, rain}) = 0.8$$

$$P(\text{ghost in } [3, 2] \mid \text{red in } [3, 2]) = 0.99$$

## Probabilistic inference

- **Evidence** variables  $\mathbf{E} = \{E_1, \dots, E_k\}$ , observed as  $\mathbf{e} = (e_1, \dots, e_k)$ .
- **Query** variable  $Q$ .
- **Hidden** variables  $\mathbf{H} = \{H_1, \dots, H_r\}$ .
- Together, they are all the variables of the model,  
 $\{Q\} \cup \mathbf{E} \cup \mathbf{H} = \{X_1, \dots, X_n\}$ .

Inference is the problem of computing the posterior distribution  $\mathbf{P}(Q \mid \mathbf{e})$ .

# Inference by enumeration

Starting from the joint distribution  $\mathbf{P}(Q, \mathbf{E}, \mathbf{H})$ , select the entries consistent with the evidence, sum the hidden variables out, and normalize:

$$\mathbf{P}(Q \mid \mathbf{e}) = \frac{1}{Z} \sum_{\mathbf{h}} \mathbf{P}(Q, \mathbf{h}, \mathbf{e}), \quad Z = \sum_q \sum_{\mathbf{h}} P(q, \mathbf{h}, \mathbf{e}).$$

**function** ENUMERATION-ASK( $Q, e, P$ ) **returns** a distribution over  $Q$

$b \leftarrow$  an empty table over the values of  $Q$

**for each** value  $q$  of  $Q$  **do**

$b[q] \leftarrow$  the sum of  $P(q, h, e)$  over the assignments  $h$

$Z \leftarrow$  the sum of  $b[q]$  over the values  $q$

**return**  $b / Z$

*Example:*

- $\mathbf{P}(W)$ ?
- $\mathbf{P}(W \mid \text{winter})$ ?
- $\mathbf{P}(W \mid \text{winter, hot})$ ?

| $S$    | $T$  | $W$  | $P$  |
|--------|------|------|------|
| summer | hot  | sun  | 0.3  |
| summer | hot  | rain | 0.05 |
| summer | cold | sun  | 0.1  |
| summer | cold | rain | 0.05 |
| winter | hot  | sun  | 0.1  |
| winter | hot  | rain | 0.05 |
| winter | cold | sun  | 0.15 |
| winter | cold | rain | 0.2  |

## Complexity

- Inference by enumeration can be used to answer probabilistic queries for **discrete variables** (i.e., with a finite number of values).
- However, enumeration **does not scale!**
  - Assume a domain described by  $n$  variables taking at most  $d$  values.
  - Space complexity:  $O(d^n)$
  - Time complexity:  $O(d^n)$

Can we reduce the size of the representation of the joint distribution?

# Independence

The joint distribution has  $d^n$  entries, but the chain rule already writes it as a product of conditional distributions,

$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i \mid x_1, \dots, x_{i-1}).$$

**Independence** and **conditional independence** state that some of these factors do not depend on all their conditions. Each dropped variable divides the size of a factor by  $d$ , which is what makes probabilistic models **scale up**.

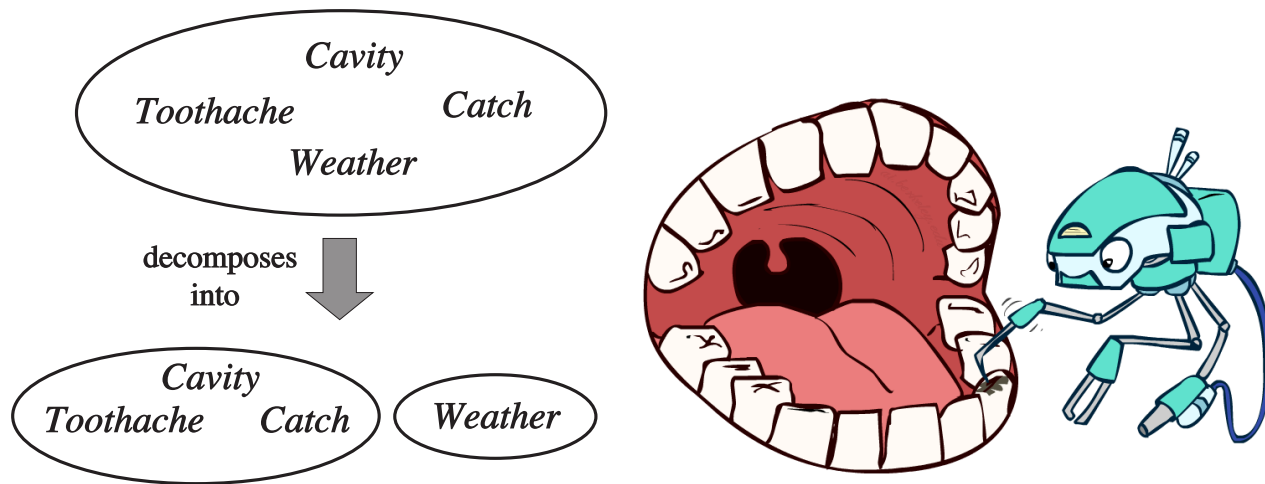
## Independence

$A$  and  $B$  are **independent**, denoted  $A \perp B$ , iff

$$P(a, b) = P(a)P(b) \quad \text{for all } a \in D_A, b \in D_B.$$

Equivalently, knowing one says nothing about the other:  $P(a \mid b) = P(a)$  whenever  $P(b) > 0$ , and  $P(b \mid a) = P(b)$  whenever  $P(a) > 0$ .

Example 1:



$$\begin{aligned} &P(\text{toothache}, \text{catch}, \text{cavity}, \text{weather}) \\ &= P(\text{toothache}, \text{catch}, \text{cavity})P(\text{weather}) \end{aligned}$$

The original 32-entry table reduces to one 8-entry and one 4-entry table (assuming 4 values for **Weather** and boolean values otherwise).

*Example 2:*

For  $n$  independent coin flips, the joint distribution can be fully **factored** and represented as the product of  $n$  1-entry tables.

- $2^n \rightarrow n$

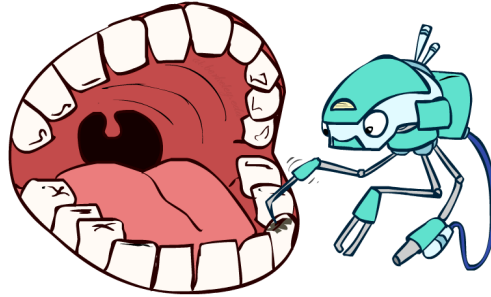
## Conditional independence

$A$  and  $B$  are **conditionally independent** given  $C$ , denoted  $A \perp B \mid C$ , iff

$$P(a, b \mid c) = P(a \mid c)P(b \mid c) \quad \text{for all } a \in D_A, b \in D_B, c \in D_C \text{ with } P(c) > 0.$$

Equivalently, once  $c$  is known, one says nothing more about the other:

$$P(a \mid b, c) = P(a \mid c) \text{ and } P(b \mid a, c) = P(b \mid c), \text{ whenever these are defined.}$$



*Example 1:*

Assume three random variables **Toothache**, **Catch** and **Cavity**.

**Catch** is conditionally independent of **Toothache**, given **Cavity**. Therefore, we can write:

$$\begin{aligned} P(\text{toothache}, \text{catch}, \text{cavity}) \\ &= P(\text{toothache} \mid \text{catch}, \text{cavity})P(\text{catch} \mid \text{cavity})P(\text{cavity}) \\ &= P(\text{toothache} \mid \text{cavity})P(\text{catch} \mid \text{cavity})P(\text{cavity}) \end{aligned}$$

In this case, the representation of the joint distribution reduces to  $2 + 2 + 1$  independent numbers (instead of  $2^n - 1$ ).

*Example 2 (Naive Bayes):*

More generally, from the product rule, we have

$$P(\text{cause}, \text{effect}_1, \dots, \text{effect}_n) = P(\text{effect}_1, \dots, \text{effect}_n \mid \text{cause})P(\text{cause})$$

Assuming the effects are **conditionally independent given the cause**, it comes:

$$P(\text{cause}, \text{effect}_1, \dots, \text{effect}_n) = P(\text{cause}) \prod_i P(\text{effect}_i \mid \text{cause})$$

This probabilistic model is called a **naive Bayes** model.

- The complexity of this model is  $O(n)$  instead of  $O(2^n)$  without the conditional independence assumptions.
- Naive Bayes can work surprisingly well in practice, even when the assumptions are wrong.

# The Bayes' rule

A model is usually given in the **causal** direction: a prior on the cause, and the distribution of its effects given the cause, as in the naive Bayes model. For the ghost and its sensors,  $\mathbf{P}(G)$  and  $\mathbf{P}(R_{i,j} \mid G)$ .

Inference asks the **diagnostic** question,  $\mathbf{P}(G \mid r_{i,j})$ : the cause, given the effects that were observed.

Bayes' rule turns one into the other, and follows from the two factorizations of the product rule.

Study the next slide. **Twice.**

# The Bayes' rule

The product rule defines two ways to factor the joint distribution of two random variables.

$$P(a, b) = P(a | b)P(b) = P(b | a)P(a)$$

Therefore,

$$P(a | b) = \frac{P(b | a)P(a)}{P(b)}.$$



- $P(a)$  is the prior belief on  $a$ .
- $P(b)$  is the probability of the evidence  $b$ .
- $P(a | b)$  is the posterior belief on  $a$ , given the evidence  $b$ .
- $P(b | a)$  is the conditional probability of  $b$  given  $a$ . Depending on the context, this term is called the likelihood.

$$P(a \mid b) = \frac{P(b \mid a)P(a)}{P(b)}$$



The Bayes' rule is the **foundation** of many AI systems.

*Example 1: diagnostic probability from causal probability.*

$$P(\text{cause} \mid \text{effect}) = \frac{P(\text{effect} \mid \text{cause})P(\text{cause})}{P(\text{effect})}$$

where

- $P(\text{effect} \mid \text{cause})$  quantifies the relationship in the **causal** direction.
- $P(\text{cause} \mid \text{effect})$  describes the **diagnostic** direction.

Let  $S$ =stiff neck and  $M$ =meningitis. Given  $P(s \mid m) = 0.7$ ,  $P(m) = 1/50000$ ,  $P(s) = 0.01$ , it comes

$$P(m \mid s) = \frac{P(s \mid m)P(m)}{P(s)} = \frac{0.7 \times 1/50000}{0.01} = 0.0014.$$

### Example 2: Ghostbusters, revisited

- A random variable  $G$  gives the ghost location, and a random variable  $R_{i,j}$  the color read at the cell  $[i, j]$ .
- We start from a uniform **prior**  $\mathbf{P}(G)$  over the locations.
- We know the sensors, i.e. the **sensor model**  $\mathbf{P}(R_{i,j} \mid G)$ .
  - e.g.,  $P(R_{1,1} = \text{yellow} \mid G = [1, 1]) = 0.1$ .
- Readings are conditionally independent given the ghost location,  
 $R_{i,j} \perp R_{i',j'} \mid G$ .

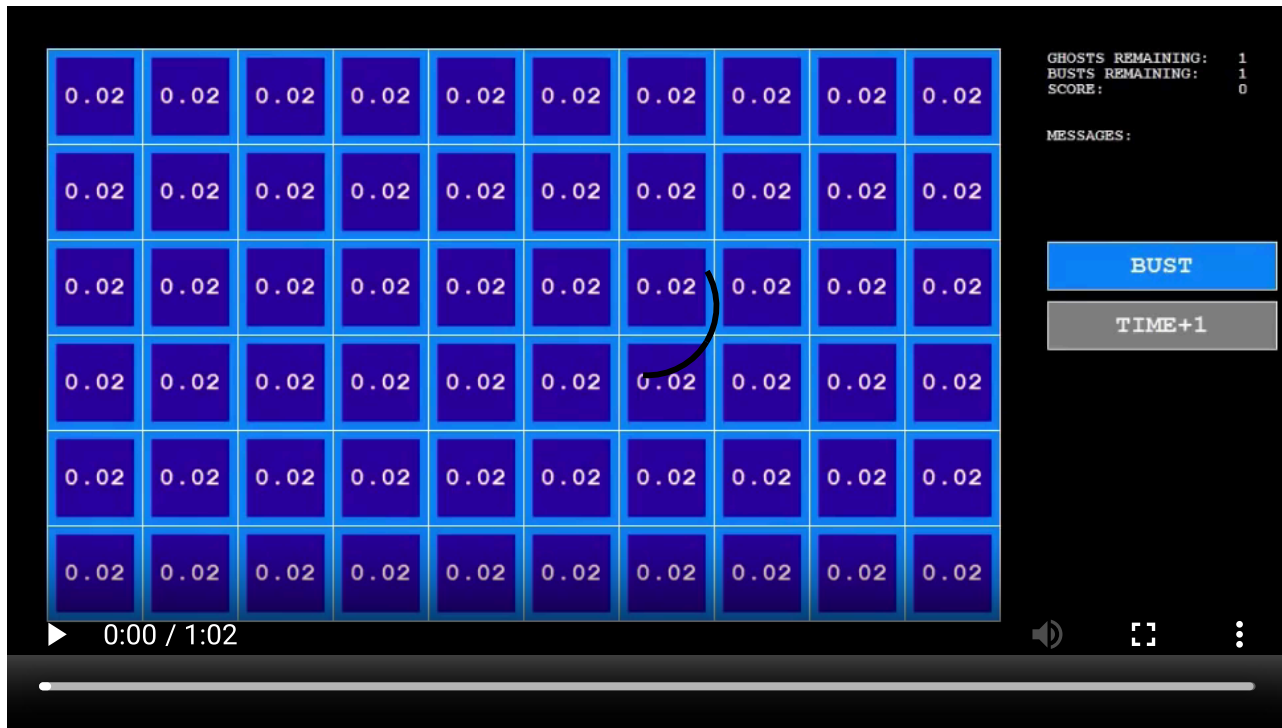
Once the reading  $r_1$  is observed, Bayes' rule gives the **posterior** over the locations,

$$\mathbf{P}(G \mid r_1) = \frac{\mathbf{P}(r_1 \mid G)\mathbf{P}(G)}{P(r_1)} = \frac{1}{Z}\mathbf{P}(r_1 \mid G)\mathbf{P}(G).$$

Each new reading updates the posterior of the previous ones, which acts as its prior,

$$\mathbf{P}(G \mid r_{1:k+1}) = \frac{1}{Z}\mathbf{P}(r_{k+1} \mid G)\mathbf{P}(G \mid r_{1:k}),$$

since the readings are conditionally independent given  $G$ .



# All the rules

## Sum rule

$$P(x) = \sum_y P(x, y)$$

## Product rule

$$P(x, y) = P(x \mid y)P(y)$$

## Chain rule

$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i \mid x_{1:i-1})$$

## Total probability

$$P(x) = \sum_y P(x \mid y)P(y)$$

## Bayes' rule

$$P(x \mid y) = \frac{P(y \mid x)P(x)}{P(y)}$$

## Normalization

$$\mathbf{P}(Q \mid \mathbf{e}) = \frac{1}{Z} \mathbf{P}(Q, \mathbf{e})$$

## Independence

$$X \perp Y \Leftrightarrow P(x, y) = P(x)P(y)$$

# Summary

- Uncertainty is **inescapable** in complex, non-deterministic or partially observable environments. Probabilities summarize what ignorance and laziness leave undecided, and measure the agent's **beliefs** rather than the world.
- A joint distribution  $\mathbf{P}(X_1, \dots, X_n)$  answers **every** query: select the entries consistent with the evidence  $\mathbf{e}$ , marginalize the hidden variables out, normalize. This is **inference by enumeration**, and it costs  $O(d^n)$ .
- **Independence** and **conditional independence** factor the joint into smaller tables, which is what makes probabilistic models scale. Naive Bayes is the extreme case,  $O(2^n)$  down to  $O(n)$ .
- **Bayes' rule**  $P(a \mid b) = P(b \mid a)P(a)/P(b)$  turns a model of the effects given the cause into a belief about the cause given the effects, and the posterior of one observation is the prior of the next.
- These beliefs are what a rational agent acts on: lecture 5 makes the computation scale, lecture 6 lets the world change with time, and lectures 8 and 9 turn posteriors into decisions.

The end.